Liu 2021 multiple myeloma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Malignant plasma cells form a per-patient clonal archipelago. Across the full θ×n_pcs Harmony sweep, 9/10 patients show self-neighbour enrichment >2.5× (worst ~63×) — cells sit next to their own clone, not a shared state. iLISI peaks at only ~59% of ceiling (θ=5) then declines; stronger integration does not merge the clones. Inter-patient clonal/CNV heterogeneity, not a sample-integrated continuum. Do not force a trajectory.

Gate 2 — module geography → NA

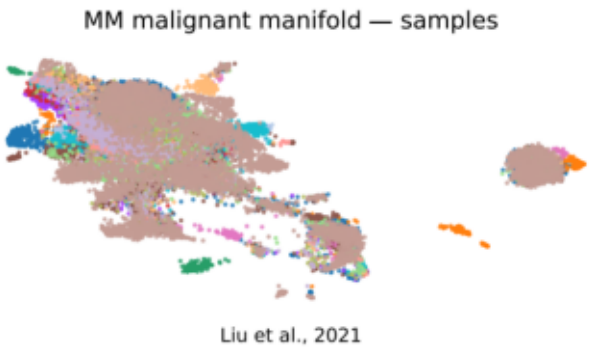
Not evaluated — Gate 1 NO-GO halts the chain before the manifold/signature steps.

Gate 3 — cross-repression → NA

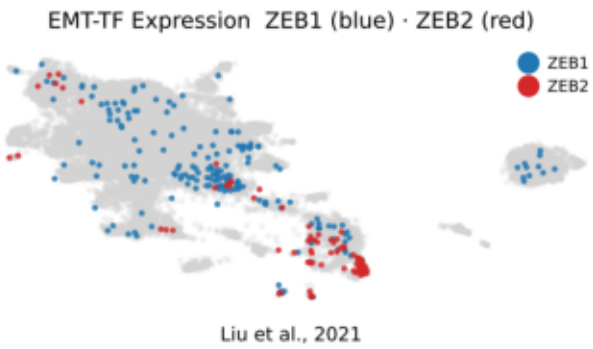
Not evaluated — branching-only, and no trajectory was fit.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
Multiple myeloma bone-marrow aspirates (Liu et al., 2021), 10x — 17,003 malignant plasma cells (15 samples ≥101 cells, 10 patients). Gate 1 NO-GO (clonal archipelago).

Verdict.
The developmental hub-and-arm pattern does NOT reproduce. Clonal MM plasma cells are irreducibly patient-segregated (Gate 1 NO-GO); no cell-state manifold or module geography can be honestly assessed. Consistent with the other MM datasets (Cohen 2021) in the atlas. EMT-TFs ZEB1 (11.1%) / ZEB2 (4.7%) — only detectable factors, descriptive only.

Parameter choices

Min-cell threshold	≥101 cells → 15/20 samples (17,003 cells); tail cliff 66→156	**Integration**	Harmony on sample; θ & n_pcs swept → θ=5, n_pcs=40 (iLISI-maximal)
Embedding	n_neighbors=30, spread=1.3	**Geometry**	none fitted (Gate 1 NO-GO); global means Linear-pSom/Crest
EMT-TFs shown	ZEB1 (11.1%) / ZEB2 (4.7%) — only detectable; VIM excluded		